

ARTIFICIAL INTELLIGENCE

CONCEPTS, TECHNIQUES,
APPLICATIONS AND ISSUES



DEFINITION OF ARTIFICIAL INTELLIGENCE



Artificial Intelligence refers to the capability of machines to imitate intelligent human behavior.

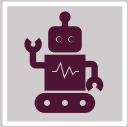


It involves learning, reasoning, perception, and decision-making.



AI aims to create rational agents that act optimally.

WHAT IS AN AI TECHNIQUE?



AI techniques are computational methods used to solve complex problems.



They rely on heuristics, learning, and approximation.



Examples: Search algorithms, Machine Learning, Neural Networks.

BASIC CONCEPTS OF AI



Intelligent
Agents



Knowledge
Representation



Reasoning and
Inference



Learning
Mechanisms



Perception and
Action

AI AGENT MODEL

Agent perceives
environment using
sensors.



Agent performs
actions using
actuators.



Rational agent
selects actions to
maximize
performance.

APPLICATIONS OF AI



Healthcare:
Diagnosis,
prediction



Finance:
Fraud
detection,
trading



Education:
Intelligent
tutoring



Robotics and
Automation



Natural
Language
Processing

CHARACTERISTICS OF AI APPLICATIONS

- Adaptability and learning
- Autonomy
- Handling uncertainty
- Data-driven behavior
- Context awareness

FEATURES OF AI SYSTEMS



Reasoning and inference



Learning and self-improvement



Pattern recognition



Problem-solving ability



Human-computer interaction

GENERAL ISSUES IN AI



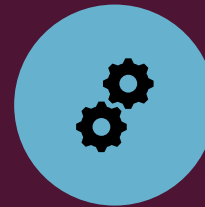
EXPLAINABILITY
AND
TRANSPARENCY



BIAS AND
FAIRNESS



PRIVACY AND
SECURITY

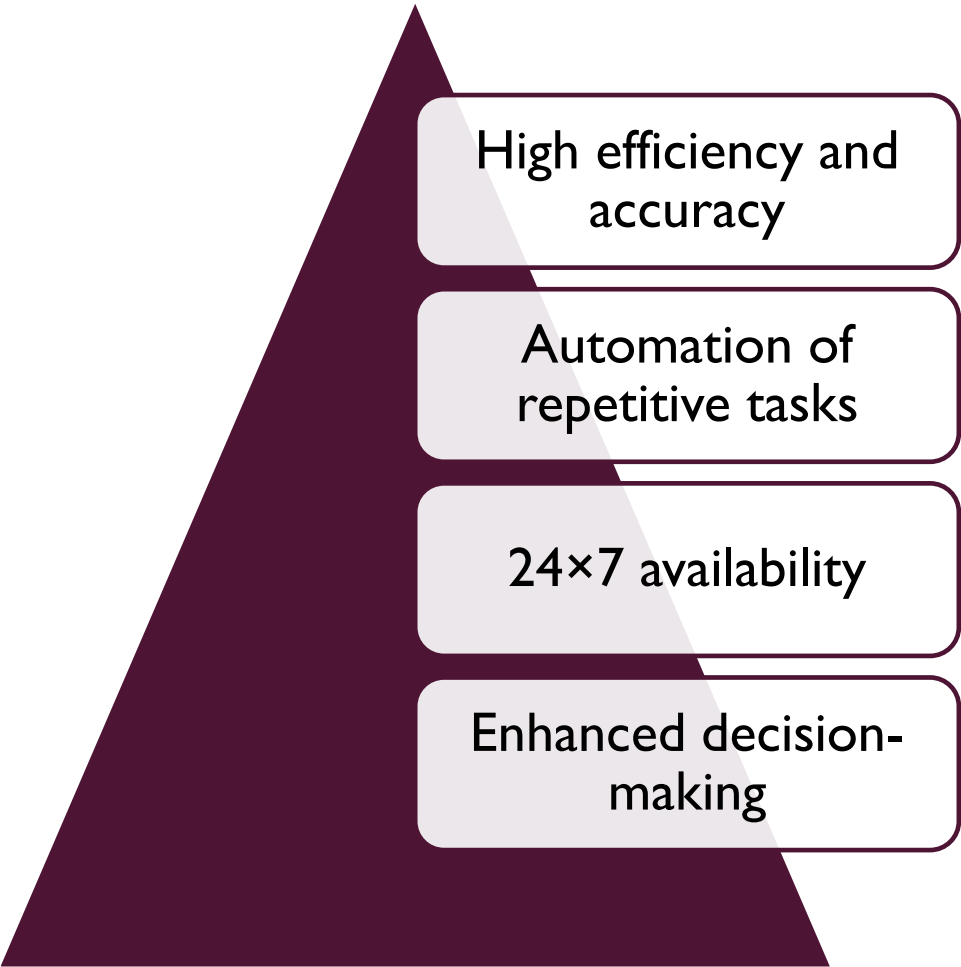


SCALABILITY AND
ROBUSTNESS



NARROW AI VS
GENERAL AI

PROS OF ARTIFICIAL INTELLIGENCE



High efficiency and
accuracy

Automation of
repetitive tasks

24×7 availability

Enhanced decision-
making

CONS OF ARTIFICIAL INTELLIGENCE



JOB DISPLACEMENT
RISKS



HIGH DEVELOPMENT
AND MAINTENANCE
COST



ETHICAL AND
SOCIAL CONCERNS



DEPENDENCE ON
QUALITY DATA

CONCLUSION

- AI is transformative but requires responsible use.
- Ethical, explainable, and human-centered AI is essential.
- Future scope depends on balanced development.

